

Bayesian Compound Identification Confidence Scoring

Dataset: LC-BinBase, HILIC negative-ion TTOF (Premier) **Confirmed spectra evaluated:** 1,631 (manually annotated, non-yy/zz) **Unconfirmed spectra expanded:** 5,899 (never previously queried) **Total library hits scored:** 195,367 candidate matches

Part I — The Problem and Approach

1.1 What We Are Solving

Standard LC-MS/MS library search returns a ranked list of candidate compounds with no uncertainty estimate. The top hit might be correct with 99% confidence, or it might be one of ten equally plausible structural isomers with no way to tell them apart. Analysts either accept all top hits blindly or spend large amounts of time manually reviewing individual spectra.

This project builds a **Bayesian evidence integration framework** that assigns each candidate compound a calibrated posterior probability:

$$P(\text{compound}_i \mid D) \propto p_i \cdot P(\text{LR}_i \mid \text{MS}_2) \cdot P(\text{LR}_i \mid \text{MS}_1) \cdot P(\text{LR}_i \mid \text{RT})$$
 An explicit novel compound hypothesis is always included so all posteriors sum to 1:

$$\sum_i P(\text{compound}_i \mid D) + P(\text{novel} \mid D) = 1$$

When no candidate is convincingly better than the novel hypothesis, the model says no. The pipeline computes InChIKey14 from SMILES for each library hit, computes it for the confirmed compound from the spectra file's SMILES, and marks a hit correct=True if they match. rather than forcing a call.

1.2 What This Is Not

- **Not a binary classifier.** There is no FP ground truth — all confirmed spectra were annotated because they had good library matches. The model has no labeled negatives. YY/ZZ could serve as fp but they are labelled not because of their fp nature but being weird as a spectra.
- **Not a re-ranking model.** Rank alone tells you nothing about the margin between candidates. The model's value is in quantifying uncertainty, not improving rank.
- **Not a general identification engine.** The model operates only on spectra that return library hits. It does not address de novo structure elucidation.

1.3 Why Bayesian

The Bayesian framework was chosen not for accuracy but for properties that matter scientifically:

1. **Calibrated output** — a posterior of 0.9 should mean the call is correct ~90% of the time
2. **Channel dropout** — when RT/predicted RT is unavailable, LR_RT = 1 and drops out silently without breaking the model
3. **Decomposable** — each channel's contribution is explicit; ablations show exactly what is and is not working

4. **Modular prior** — the prior can be swapped or improved independently of the likelihood terms
 5. **Directly interpretable** — $P(\text{threonine} \mid \text{this spectrum}) = 0.976$ is a statement that anyone could comprehend.
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Part II — Data

2.1 Spectra

ttof+neg+hilic.csv — 7,530 LC-MS/MS features from a HILIC negative-ion TTOF (Premier) run processed through LC-BinBase.

Key fields used: **wiki_id**, **rt** (observed retention time, seconds), **precursor_mz**, **entropy** (spectral entropy H of the MS2 spectrum).

2.2 Library Hits — Two Sources

All library hits were fetched from MassWiki, which aggregates public and in-house spectral libraries.

Reference library (public: NIST23, GNPS, MassBank)

- Broad chemical coverage, big, variable spectral quality
- RT channel: Kong 2.0 predicted RT (**predicted_rt_hilic**)
- Mean entropy similarity on confirmed correct hits: 0.798

Annotation library (in-house)

- Spectra measured on the same instrument (under the same conditions like collision energy?)
- RT channel: real measured RT offset (**anno_delta_rt** = observed RT – library RT)
- Mean entropy similarity on confirmed correct hits: 0.989
- Higher similarity and tighter RT

When an annotation hit exists, it typically dominates. The two sources receive separate RT likelihood parameters ($\sigma_{\text{RT_anno}} = 14.1$ s vs $\sigma_{\text{RT_ref}} = 19.1$ s) reflecting this quality difference.

2.3 Biological Prior

Built from 17,415 confirmed TTOF HILIC negative records in **hilic_ttof_all.csv**, queried from BinBase. Each compound's prior weight is:

$$\pi_{\text{raw}} = \left(\sum_{\text{records}} \text{peak_gaussian_similarity} \times \text{peak_purity} \right) \times \sqrt{N_{\text{studies}}}$$

Quality-weighting by peak shape and purity downweights poorly resolved peaks; the square root of study count rewards compounds confirmed across multiple independent studies.

Part III — The Scoring Model

3.1 MS2 Likelihood Ratio

Entropy similarity (0–1) is converted to a likelihood ratio via softmax, weighted by spectral complexity:

$$\lambda(H) = 1 - e^{-\alpha H}$$

$$\text{LR}_{\text{MS2},i} = \left[\lambda \cdot \frac{e^{s_i}}{\sum_j e^{s_j}} + (1 - \lambda) \cdot \frac{1}{N} \right] \times N$$

For $H \approx 0$ (single-ion spectrum), $\lambda \approx 0$ — MS2 is uninformative, the model falls back to the prior. For $H > 0.5$, $\lambda \approx 1$ — MS2 is fully trusted. The parameter $\alpha = 10.0$ was estimated by maximum likelihood on confirmed spectra; it consistently hits the upper bound. Only 134 / 1,631 (8.2%) of confirmed spectra have $H < 0.5$, suggesting that the complexity weighting is only meaningfully active for the 134 spectra below that threshold. Even at $H = 0.3$, MS2 is still 95% weighted.

3.2 MS1 Likelihood Ratio

Gaussian on Δppm with $\sigma_M = 4.21$ ppm fitted from confirmed correct hits. The denominator represents a random unrelated compound (null mass distribution, $\sigma = 50$ ppm):

$$\text{LR}_{\text{MS1},i} = \frac{\mathcal{N}(\Delta\text{ppm}_i; 0, 4.21)}{\mathcal{N}(0; 0, 50)}$$

Note: TTOF is less mass-accurate than Orbitrap ($\sigma_M \approx 4.2$ ppm vs 1–2 ppm). MassWiki pre-filters candidates by precursor m/z , so most candidates for a given spectrum already have small Δppm values — the MS1 channel is most valuable when a candidate has an unusually large error (e.g. the 14.6 ppm hit in Example 2, Section 5.2).

3.3 RT Likelihood Ratio — Two Channels

Annotation hits (real measured RT):

$$\text{LR}_{\text{RT},i} = \frac{\mathcal{N}(\text{anno_delta_rt}_i; 0, 14.1, \text{s})}{\mathcal{N}(0; 0, 300, \text{s})}$$

Reference hits (Kong predicted RT): $\text{LR}_{\text{RT},i} = \frac{\mathcal{N}(RT_{\text{obs}} - \widehat{RT}_i; 0, 19.1, \text{s})}{\mathcal{N}(0; 0, 300, \text{s})}$

If no RT is available, $\text{LR}_{\text{RT}} = 1$ (channel drops out). The 300 s null represents a compound eluting at a random position on the gradient for a 5 min method (which is what we are using I suppose?).

3.4 Novel Compound Term and Abstention

$$P(\text{novel} \mid D) = \frac{P_{\text{novel}}}{\sum_i P_{\text{unnorm},i} + P_{\text{novel}}}$$

If $P(\text{novel} \mid D) \geq 0.6$, the model abstains. $P_{\text{novel}} = 0.224$ is estimated from the fraction of all 7,530 MassWiki download spectra with no library hit.

3.5 Fitted Parameters

All parameters estimated from the 1,631 confirmed, human annotated spectra:

Parameter	Value	Method
σ_M	4.21 ppm	Trimmed std of Δppm on InChIKey14-matched correct hits

Parameter	Value	Method
$\sigma_{\text{RT_anno}}$	14.1 s	Std of <code>anno_delta_rt</code> on confirmed annotation hits, trimmed ± 60 s
$\sigma_{\text{RT_ref}}$	19.1 s	Std of <code>obs_RT</code> – <code>predicted_RT</code> on confirmed reference hits, trimmed ± 120 s
α	10.0	MLE on confirmed spectra (hits optimizer upper bound)
P_{novel}	0.224	Fraction of all 7,530 spectra with no library hit

Part IV — Performance on Confirmed Spectra

4.1 Overall Accuracy

Dataset: 1,631 confirmed spectra (correct compound present in library for all).

Method	Top-1 Accuracy
Raw entropy similarity rank	99.8%
Full Bayesian model	90.3%
LOO cross-validation	90.7%
Abstentions	22 / 1,631 (1.3%)

The 9.5 pp drop from the entropy baseline is expected and not a failure. The confirmed set was annotated via library search — entropy similarity nearly always ranks the correct compound first by construction. The RT and prior channels introduce noise on this set because the prior is nearly flat across most candidates and σ_{RT} (14–19 s) is wide relative to RT differences between structural isomers. The baseline's 99.8% is a ceiling artifact, not a true performance benchmark.

4.2 Calibration

The model's primary contribution is not re-ranking but calibrated uncertainty. When the model is confident, it is right:

Posterior bin	N spectra	Observed accuracy
> 0.90	93	100%
0.70–0.90	145	99.3%
0.50–0.70	253	97.2%
0.30–0.50	418	93.3%
< 0.30	700	83.9%

The monotonic relationship between posterior and accuracy is what makes the score actionable. A threshold of 0.70 gives 99%+ precision on this dataset. How we generalize this to other datasets or other methods remains unknown.

4.3 The Duplicate Entry Problem

87.9% of confirmed spectra have more than one correct-labeled library entry (mean 6.0 per spectrum). The same compound appears under multiple names, — all sharing the same InChIKey14. The model scores these as independent candidates, splitting posterior mass.

Example: Threonine has 41 correct entries across the library. The top posterior per entry is 0.063 — which looks like low confidence — but the sum across all threonine entries is 0.98.

InChIKey14 deduplication before scoring collapses all entries with the same molecular connectivity into one candidate before scoring would eliminate posterior dilution. The pipeline computes InChIKey14 from SMILES for each library hit, computes it for the confirmed compound from the spectra file's SMILES, and marks a hit `correct=True` if they match.

4.4 Failure Analysis

159 of 1,631 confirmed spectra are called incorrectly. Of these:

- **68 (43%)**: correct compound is rank 2 — near-tie, median posterior gap = 0.084
- **91 (57%)**: correct compound is rank 3 or lower

Representative failure patterns:

Name fragmentation (a6ICAHB/4704): The correct compound Fuc(a1-2)Gal(b1-4)GlcNAc has perfect entropy similarity (1.000) but its probability is split with synonyms. Lewis A trisaccharide — a different name for essentially the same compound — takes the top posterior (0.775) while the correctly-named entry scores 0.096.

Lipid notation ambiguity (a6ICAHB/4072): PE O-18:1_16:1 and PE O-16:1_18:1 are sn-positional isomers of the same ether-linked phosphatidylethanolamine. The model calls PE O-18:1_16:1 (wrong sn-position assignment) at 0.645. The correct entry PE O-16:1_18:1 scores 0.103. MS2 cannot distinguish sn-positions — RT or additional fragmentation would be needed.

Synonym conflict (a6ICAHB/900): 1,2-Dioleoyl-sn-glycero-3-phosphate (wrong) scores 0.576 while its synonym 1-Oleoyl-L- α -lysophosphatidic acid (correct) scores 0.081 and LPA 18:1 (correct) scores 0.013. All three refer to the same compound.

Part V — Expansion to Unconfirmed Spectra

5.1 Motivation

The initial confirmed pipeline is inherently circular: fitted on confirmed spectra, evaluated on confirmed spectra, which were themselves selected because they had good library matches. The 5,899 unconfirmed spectra had never been queried against MassWiki. Expanding the search:

1. Provides a genuine test of the model on spectra selected independently of library search quality
2. Generates a prioritised annotation queue to replace arbitrary manual curation
3. Establishes the true `P_novel` across the full run

5.2 Library Hit Coverage — Correcting `P_novel`

Category	N	% of 7,530 total
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Category	N	% of 7,530 total
Confirmed spectra	1,631	21.7%
Unconfirmed with library hits	4,216	56.0%
True no-hit (no library match)	1,683	22.4%

True P_{novel} = 0.224. The previous estimate of 0.783 was entirely circular — it counted unqueried spectra as "no hit." Most unconfirmed spectra do have library candidates; they had simply never been scored.

5.3 Scoring Results

167,857 candidate hits across 4,216 spectra were scored using the same parameters fitted on the confirmed set (mean 39.8 candidates per spectrum vs 17.5 for confirmed spectra — the unconfirmed set has broader and noisier library matches).

Outcome	N	%
Called ($P_{\text{novel}} < 0.6$)	3,119	74.0%
Abstained ($P_{\text{novel}} \geq 0.6$)	1,097	26.0%

Confidence distribution of called spectra:

Posterior bin	N
> 0.90	110
0.70–0.90	82
0.50–0.70	117
0.30–0.50	327
< 0.30	2,483

The distribution is heavily skewed toward low confidence. The unconfirmed set contains many genuinely ambiguous cases with no annotation to guide selection. However, we identified 190 high confidence calls that were not previously annotated for analytical chemists to review.

5.4 Review Queue

2,253 spectra flagged across four categories for chemist review:

Flag 1 — HIGH_CONF_NOVEL (156 spectra)

Posterior > 0.70 on a compound not previously confirmed in BinBase

wiki_id	Compound	Posterior	H	Entropy sim	Δppm	Source
a6ICAHB/6660	Thr-Asp	0.999	1.23	0.026	−1.71	annotation

wiki_id	Compound	Posterior	H	Entropy sim	Δ ppm	Source
a6ICAHB/4929	D(+)-Trehalose	0.999	2.13	0.263	-0.41	reference
a6ICAHB/2469	Atranone A	0.999	2.17	0.055	+1.87	annotation
a6ICAHB/5767	Glycodeoxycholic acid 3-sulfate	0.999	0.14	0.344	-3.02	annotation
a6ICAHB/243	PC 18:0	0.998	1.21	0.327	-2.88	annotation

These are the highest-priority review candidates. If RT is chemically plausible, each represents a new confirmed annotation. Note that several entries have low entropy similarity (e.g. Thr-Asp: 0.026) — confidence is driven by RT and MS1 alignment rather than spectral matching. These should be verified with extra care.

Flag 2 — NEAR_TIE (1,188 spectra)

Top two candidates within 0.05 posterior of each other, top posterior > 0.10

The model cannot resolve between two candidates. The most common cause is structural isomers or lipid species with similar masses and RT predictions. An authentic standard or manual RT inspection would resolve most cases. This list is most useful for **prioritising standard purchases** — buy standards for the compounds that appear most frequently in near-tie spectra.

Flag 3 — RICH_NO_HIT (845 spectra)

Spectral entropy $H > 1.0$, model abstains

Well-fragmented spectra (median $H = 1.65$, max $H = 3.55$) with no convincing library match. These are the most scientifically interesting for discovery — the spectrum contains real chemical information but nothing in MassWiki explains it.

Example: [a6ICAHB/4373](#) has $H = 3.547$ (extremely information-rich) but $P_{\text{novel}} = 0.999$. The best library candidate (a pyrimido-benzazepine derivative) has entropy similarity 0.178 and a 98 s RT mismatch — clearly wrong. This spectrum is a strong candidate for de novo structure elucidation.

Flag 4 — MS1_RESCUE (64 spectra)

Entropy similarity < 0.50 but posterior > 0.60

RT and mass accuracy are doing the heavy lifting; spectral matching is poor. These calls are the most suspicious and warrant explicit verification.

Example: [a6ICAHB/4008](#) is called as γ -Glu-Leu at posterior 0.997 with entropy similarity of only 0.051. The near-zero mass error (-0.14 ppm) and annotation RT match drive the call. A 23 s Δ RT on an annotation hit is within the fitted $\sigma_{\text{RT_anno}} = 14.1$ s window but is on the wide side — the RT trace should be inspected.

Part VI — Limitations and Known Issues

6.1 Confirmed Set Bias

All parameters (σ_M , σ_{RT} , α) are fitted on confirmed spectra, which were confirmed because they had good library matches. This biases:

- α toward the spectral quality of curated spectra — α always hits its upper bound because confirmed spectra almost always have good MS2 signal
- The training set has no genuine FP ground truth
- How can we make sure the confirmed spectra are "confirmed" enough that no errors are within this set?

6.2 Annotation Library RT Offset?

$\sigma_{RT_anno} = 14.1$ s is wider than expected for same-instrument measurements, and the median annotation ΔRT is +9.2 s may suggest a systematic shift? (This does not make sense to me as those rts are corrected via istds). This may indicate a run-to-run RT drift or gradient calibration difference between the annotation library and the current run. The Gaussian model assumes zero-mean error; the actual distribution is biased.

6.3 Duplicate Library Entries

As described in Section 4.3, 87.9% of confirmed spectra have multiple entries for the same compound. This dilutes posteriors and artificially lowers per-entry confidence. InChIKey14 deduplication before scoring would be helpful without any new data.

6.4 α Is Unidentifiable

α always hits the upper bound (10.0) in every LOO (leave one out) fold. The confirmed spectra provide no leverage to estimate spectral complexity weighting because nearly all confirmed spectra have $H > 0.5$ (where $\lambda \approx 1$ regardless of α). α is effectively fixed, not learned.

6.5 Initial P_{novel} Was Circular

The original $P_{\text{novel}} = 0.783$ was computed from the fraction of spectra with no library hit, but only confirmed spectra had been queried. Unconfirmed spectra were guaranteed to have no hits. The correct $P_{\text{novel}} = 0.224$ substantially changes the abstention behavior — with a lower novel term, weak library evidence is less easily suppressed.

6.6 High-Confidence Wrong Calls in Unconfirmed Set

Some HIGH_CONF_NOVEL calls have high posterior driven by RT/MS1 alone with very low entropy similarity. Without a confirmed annotation or authentic standard, these cannot be verified. Any automated acceptance of these calls would require an independent check (e.g. RT consistency with the compound's chemical class on HILIC, or cross-referencing against another study).

Part VII — Recommended Use

Confirmed spectra

Posterior	Action
> 0.90	Accept identification (100% precision on this dataset)
0.70–0.90	Accept with note for review
0.50–0.70	Report tentative; include top 2–3 candidates
< 0.30 or abstain	Report as unidentified

Unconfirmed spectra review queue

Priority	Flag	N	Action
1	HIGH_CONF_NOVEL, post > 0.90	110	Verify RT; if plausible, accept as new annotation
2	MS1_RESCUE	64	Inspect RT trace; low entropy sim calls need verification
3	RICH_NO_HIT, H > 2.0	~200	De novo elucidation queue
4	NEAR_TIE	1,188	Prioritise authentic standard purchases

Part VIII — Proposed Next Steps

Improvement	Expected impact
InChIKey14 deduplication before scoring	Eliminates posterior dilution; likely pushes confirmed accuracy above 95%
Correct/address annotation library RT offset (+9.2 s systematic shift)	Fixes suppression of correct annotation hits; most impactful for same-platform identifications
Fetch all 7,530 spectra (not just annotated) for parameter fitting? maybe? still thinking about it	
Library source quality weighting/biological plausibility weighting (how often is this chemical detected in the same matrix accross studies? /chemical plausibility weighting (pubmed/patent count)	Better priors
SIRIUS fingerprint as additional channel	Independent spectral-to-structure prediction; genuinely orthogonal to entropy similarity
Mixture model on full score distribution	Learn TP/FP distributions from all spectra, not just confirmed ones; no FP ground truth needed

Appendix — Output Files

File	Description	Rows
out_proposal/assertions.csv	Per-(spectrum, candidate) posteriors, confirmed set	28,510
out_proposal/top_calls.csv	Top call per confirmed spectrum	1,631
out_unconfirmed/unconfirmed_scored.csv	Per-(spectrum, candidate) posteriors, unconfirmed set	167,857
out_unconfirmed/unconfirmed_top_calls.csv	Top call per unconfirmed spectrum	4,216
out_unconfirmed/review_queue.csv	Flagged spectra for chemist review	2,253
data/hilic_ttof_neg_masswiki_hits_new.csv	Library hits for confirmed spectra	120,048
data/hilic_ttof_neg_masswiki_hits_unconfirmed.csv	Library hits for unconfirmed spectra	167,857